

Pseudocode to Define Buildable Area

Labelling Boundary Lines for Parcels of Any Shape

Description:

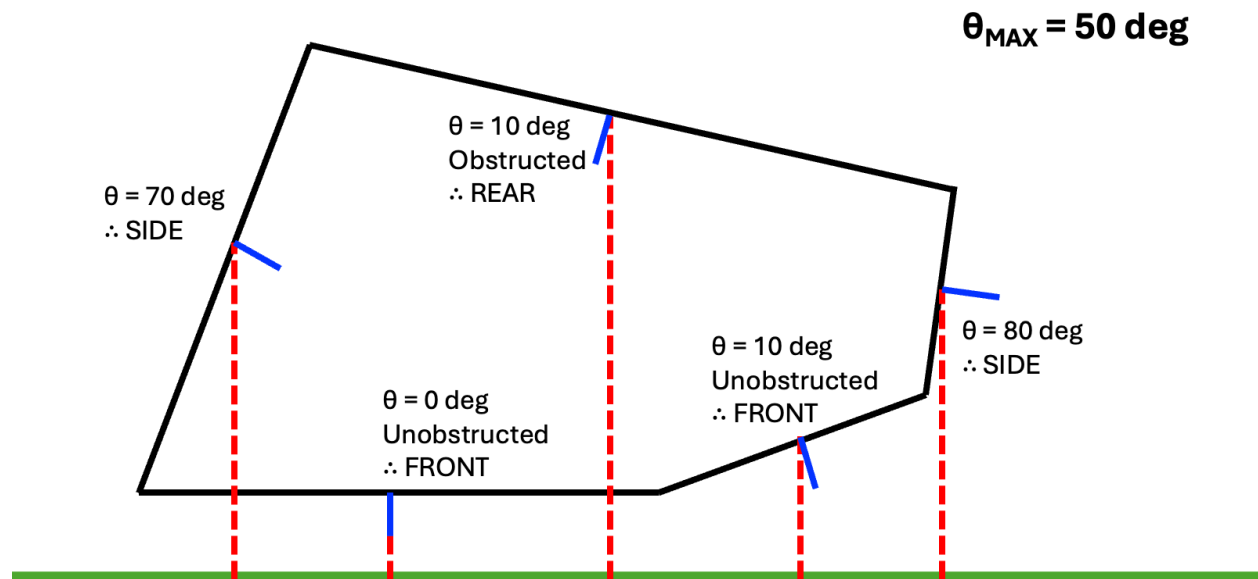
This algorithm labels each boundary line of a parcel as FRONT, SIDE, or REAR. Broadly speaking, labels are assigned based on the degree to which the boundary faces the parcel's street.

Dependencies:

- Parcels
- Streets

Steps:

1. Extract parcel perimeters as continuous boundary lines
2. Explode the parcel perimeters into individual boundary lines
3. For each boundary line, determine the acute angle between the line's perpendicular bisector and the shortest line between the boundary line's midpoint and its street (aka: bee-line).
 - a. If the angle is below θ_{MAX} , a threshold to be fine-tuned (eg: 50 deg):
 - i. If the bee-line does not intersect any boundary lines (including those from other parcels), the boundary line is FRONT
 - ii. If the bee-line intersects a boundary line from the same parcel, the boundary line is REAR
 - b. Else, the boundary line is SIDE



Applying Lot Boundary Setbacks

Description:

This algorithm applies a setback to each boundary line, and the remaining space in the parcel is the buildable area ignoring existing structures.

Dependencies:

- Labelled boundary lines
- Streets
- Lanes

Steps:

1. If the boundary line is FRONT, apply **Street Non-Flanking** setback
2. If the boundary line is SIDE:
 - a. If the boundary line overlaps another parcel, apply **Internal Side** setback
 - b. Else:
 1. If the boundary line's midpoint is closer to a street than a lane, apply **Street Flanking** setback
 2. Else, apply **Internal Side** setback
3. If the boundary line is REAR:
 - a. If the boundary line overlaps another parcel, apply **Internal Rear** setback
 - b. Else:
 1. If the boundary line's midpoint is closer to a street than a lane, apply **Street Flanking** setback
 2. Else, apply **Lane** setback

Subtracting Existing Structures

Description:

This algorithm finds buildable areas while considering existing structures. The demolition of accessory structures is assumed to be allowable.

Dependencies:

- Buildable areas without existing-structure considerations
- Buildings

Steps:

1. Label the largest building on each parcel as **Principal Front**, all others as **Accessory**
2. Apply a buffer to the **Principal Front** based on the minimum front-rear principal distance
3. Subtract the **Principal Front** (with buffer) from the buildable area to account for existing structures